

ANALYSIS REPORT – G3

MISSION 4

Analysis Report by Joan Drewitt & Rani Nalam

Topic: Analyzing Dataset from gaming company with SQL queries and results

(Task 3: Question 1- Total Revenue) by Joan Drewitt

Objective: -

Purpose of Analysis:

The purpose of this analysis is to calculate the total revenue generated by each game and identify the top-performing games based on revenue. This analysis helps which games, and genre performed the best and which year made the biggest contribution to the company's revenue.

Context:

Revenue is one of the most important ways to measure how well a business is performing. Calculating the total revenue generated by games/genre helps identify the best-selling games, understand sales patterns, and support decisions on future business growth.

Joining Method: -

Join Type and Key Fields:

An INNER JOIN was used between the **games_revenue** table and the **games_description** table using the common field **game_id**.

Used inner joins for the below questions in Task 3:

- What are the top 5 games by revenue.
- Identify the top 3 genres by total revenue.
- Which year generated most of the revenue in the company.

SQL Statement for question “What are the top 5 games by revenue”:

```
SELECT gd.game_name, gr.game_id, gr.number_of_purchases, gr.unit_price,  
gr.Number_of_Purchases * gr.Unit_Price AS Total_Revenue  
FROM games_revenue gr  
INNER JOIN games_description gd  
on gr.game_id = gd.game_id  
ORDER BY Total_Revenue desc  
LIMIT 5;
```

Rationale:

The **games_revenue** table contains purchase and pricing information, while the **games_description** table contains game names and descriptive details. Using an INNER JOIN ensured that only games with matching records in both tables were included in the analysis, providing accurate revenue calculations.

Insight Explanation: -

Insights:

➤ Description:

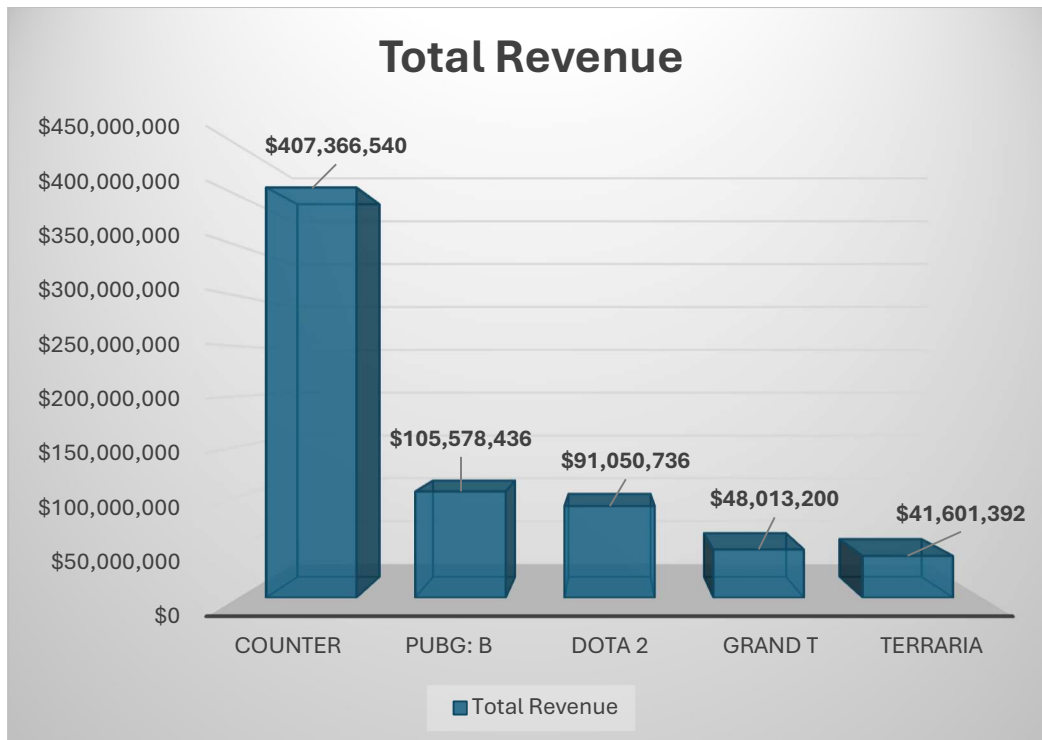
The revenue analysis identified the games, genres, and release year that generated the highest revenue within the dataset. The results provide an overview of product performance and revenue trends across different categories.

Revenue was calculated using the formula:

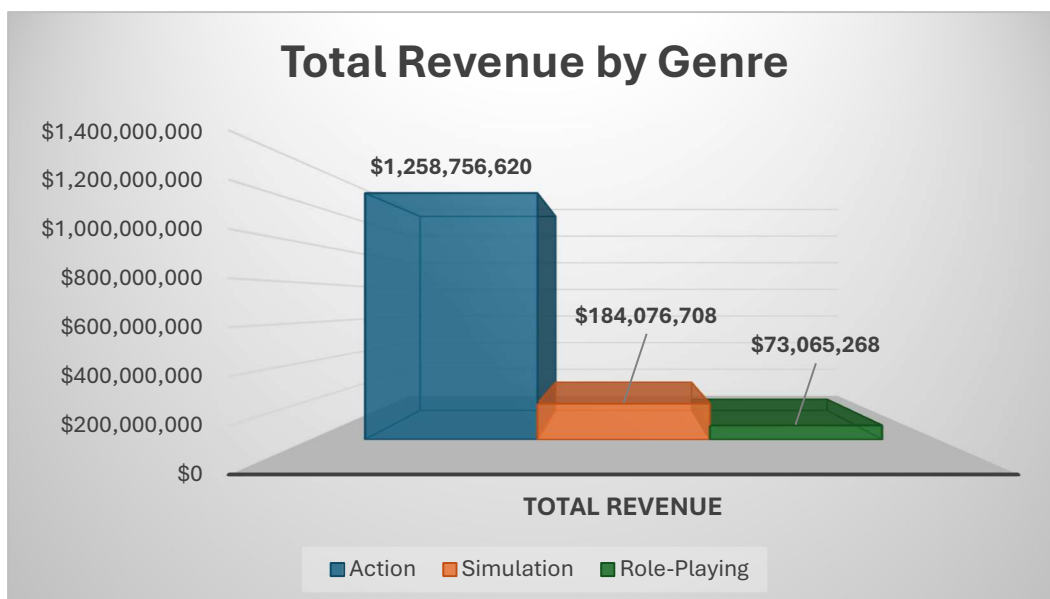
$$\text{Total Revenue} = \text{Number_of_Purchases} \times \text{Unit_Price}$$

➤ Supporting Data:

Analysis based on revenue for **top 5 games**, where “Counter” generated the highest revenue among all games, with a total revenue of **\$407,366,540** followed by PUBG: B, Dota 2, Grand T and Terraria. The findings show that these games were key revenue drivers and performed exceptionally well in the market.



At the genre level, **Action** generated the highest total revenue (**\$1,258,756,620**), followed by **Simulation** (**\$184,076,708**) and **Role-Playing** (**\$73,065,268**). The results indicate that Action games were the most profitable genre in the dataset, generating significantly more revenue than the other top genres



Conclusion:-

Summary of Key Findings:

The revenue analysis showed which **games, genres, and release year** generated the most revenue. Among all games, **Counter** generated the highest revenue of **\$407,366,540**, making it the top-performing game. The analysis also found that the **Action** genre generated the highest total revenue of **\$1,258,756,620**, which was much higher than the other top genres. In addition, **2012** was the year with the highest revenue, generating a total of **\$427,990,780**.

Recommendations:

Based on the findings company should consider investing more on high revenue generated genres such as **Action and Stimulation** on future projects and analyse more based on likes and reviews on this genre. Also need to analysis could be conducted to find the strategies and features of the game "**Counter**" which generated highest revenue among all games.

In addition, need a detailed analysis on the high-performing year 2012 which helps the company to improve their future game development and marketing decisions.

Limitations:

Analysis is only based on number of purchase and unit price so analysis may change with total cost and factors involved.

Task 3: Question 2 & 3 – Review Percentage/English Review Analysis by Rani Nalam

Q2 – Assessment: Review Percentage

Purpose of the analysis:

The goal of this analysis is to find out what percentage of customers reviewed a game after buying it.

This is helpful to gauge customer engagement and desire to provide feedback post-purchase.

Context:

Customer reviews play an important role in influencing buying decisions and providing valuable feedback to developers. Knowing what percent of buyers leave a review helps gauge user engagement and the effectiveness of review systems.

2. Joining Method

Join Type and Key Fields:

The game_id field was used to INNER JOIN the games_revenue and games_reviews tables.

```
SELECT * FROM games_revenue revenue INNER JOIN games_reviews reviews ON  
revenue.game_id =
```

```
reviews.game_id
```

Rationale:

The INNER JOIN will make sure that only games that are present in both datasets are included in the analysis. This allows us to compare purchase data against review data for the same games, allowing us to accurately calculate the percentage of reviews.

3. Insight explanation

The analysis is the percentage of customers that left the review against the total number of purchases.

Supporting Data

The percentage was calculated as follows:

$\text{ROUND}(\text{SUM}(\text{games_reviews.number_of_reviews_from_purchased_people}) * 100 / \text{SUM}(\text{games_revenue.Number_of_Purchases}), 2)$

This formula compares the total number of reviews from purchasers with the total number of purchases of all games.

Relevance:

The higher the percentage the more engaged the customers are, the more they participate in providing feedback. That might mean customers are less likely to leave reviews after buying games, which could be why the percentage is lower.

4. Conclusion

Summary of the Key Findings:

The analysis found out the percentage of buyers who wrote reviews after buying games. And this metric provides an idea of overall levels of customer engagement

Recommendations:

- Encourage and incentivize more customers to leave reviews.
- Emphasize the role of reviews in enhancing game quality.
- * Monitor trends in the participation in reviews over time.

Limitations:

- The analysis assumes that all reviews were collected from verified buyers.
- It doesn't consider the sentiment or quality of reviews.

Q3 – Assessment: English Review

1.Objectives: -

Purpose of the Analysis:

The aim of this analysis is to find out what percentage of reviews are written in English for each game.

Context:

Knowing the language origin of reviews can help determine each game's target audience and can help with marketing and localization.

2. Joining Method: -

Type and Key Fields Join:

INNER JOIN between the tables games_reviews and games_description on the field game_id
FROM games_reviews gr INNER JOIN games_description gd ON gr.game_id = gd.game_id

Reasoning:

The join enables to associate review statistics with game names, and consequently to interpret and present the results more easily.

3.Insight Explanation: -

Description:

The analysis computes the share of reviews conducted in the English language for each game.

Supporting Data:

```
ROUND(  
gr.number_of_english_reviews * 100.0 /  
gr.number_of_reviews_from_purchased_people,2  
) AS english_review_percentage
```

This computation scopes the reviews in the English language against the total purchaser reviews of each game.

Relevance:

If a game reports a higher share of English reviews, it may primarily cater to English-speaking customers. Alternatively, if a game reports a lower share of English reviews, it may cater to customers in diverse geography.

Example Insight:

- Games whose English review share is close to 100% are most likely reviewed by Englishspeaking users.
- Games with a lower share of English reviews most likely have a diverse user base.

4. Conclusion: -

Summary of Key Findings:

This study determined the percentage of English reviews per game and, consequently, the English makeup of the player base.

Recommendations:

- Target English-language marketing resources on games with English-speaking player bases.
- Identify games with review languages diversity for localization and multilingual services.
- Analyse review language consistently for other market trends.

Limitations:

- Only English reviews and total review counts were considered.
- Reviews in other languages were not considered.
- No review content or sentiment analysis was conducted